Fuzzy Controller for the sprinkler system,

here is a detailed explanation of the code:

In [1]:
```
#!pip install scikit-fuzzy
```

In [2]:
```
import numpy as np
import skfuzzy as fuzz
from skfuzzy import control as ctrl
```

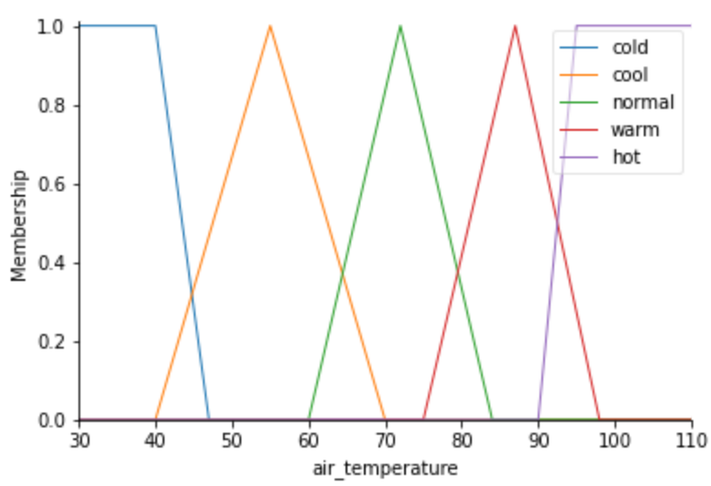
In [3]:
```
# Define antecedents (inputs)
air_temperature = ctrl.Antecedent(np.arange(30, 111,1), 'air_temperature')
soil_moisture = ctrl.Antecedent(np.arange(5, 31,1), 'soil_moisture')
```

In [4]:
```
# Define consequents (outputs)
watering_duration = ctrl.Consequent(np.arange(5, 61,1), 'watering_duration')
```

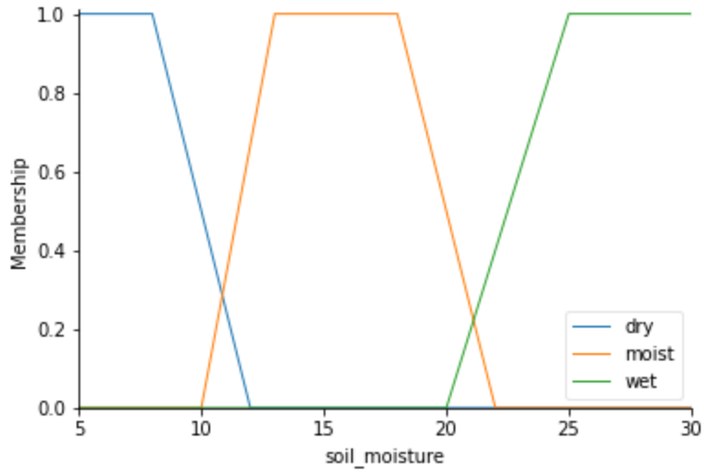
In [5]:
```
# Auto-membership function population is possible with .automf(3, 5, or 7)
#air_temperature.automf(5)
#soil_moisture.automf(3)
```

In [6]:
```
# Define fuzzy membership functions

# Set the membership functions for the input variables
air_temperature['cold'] = fuzz.trapmf(air_temperature.universe, [30, 30, 40, 47]
air_temperature['cool'] = fuzz.trimf(air_temperature.universe, [40, 55, 70]) # tr
air_temperature['normal'] = fuzz.trimf(air_temperature.universe, [60, 72, 84])
air_temperature['warm'] = fuzz.trimf(air_temperature.universe, [75, 87, 98])
air_temperature['hot'] = fuzz.trapmf(air_temperature.universe, [90, 95, 110, 110

soil_moisture['dry'] = fuzz.trapmf(soil_moisture.universe, [5, 5, 8, 12])
soil_moisture['moist'] = fuzz.trapmf(soil_moisture.universe, [10, 13, 18, 22])
soil_moisture['wet'] = fuzz.trapmf(soil_moisture.universe, [20, 25, 30, 30])

# Set the membership functions for the output variable
watering_duration['short'] = fuzz.trapmf(watering_duration.universe, [5, 5, 10, 2
watering_duration['medium'] = fuzz.trapmf(watering_duration.universe, [10, 20, 40
watering_duration['long'] = fuzz.trapmf(watering_duration.universe, [40, 45, 60,
```

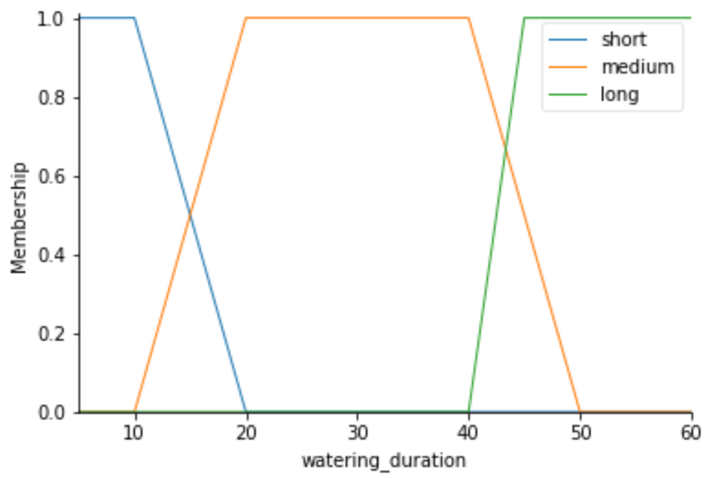
In [7]:
```
import warnings
warnings.filterwarnings('ignore', category=UserWarning)
air_temperature.view()
```



In [8]:
```
soil_moisture.view()
```



In [9]:
```
watering_duration.view()
```



In [10]:
```
# Define fuzzy rules
rule1 = ctrl.Rule(air_temperature['hot'] & soil_moisture['dry'], watering_duratio
rule2 = ctrl.Rule(air_temperature['warm'] & soil_moisture['moist'], watering_dura
rule3 = ctrl.Rule(air_temperature['warm'] & soil_moisture['dry'], watering_durati
rule4 = ctrl.Rule(air_temperature['hot'] & soil_moisture['moist'], watering_durat
```

In [11]:
```
# Create fuzzy control system
sprinkler_system = ctrl.ControlSystem([rule1, rule2, rule3, rule4])
```

In [12]:
```
# Create fuzzy simulator
sprinkler_simulator = ctrl.ControlSystemSimulation(sprinkler_system)
```

In [13]:
```
# Set input values
sprinkler_simulator.input['air_temperature'] = 92
sprinkler_simulator.input['soil_moisture'] = 11
```

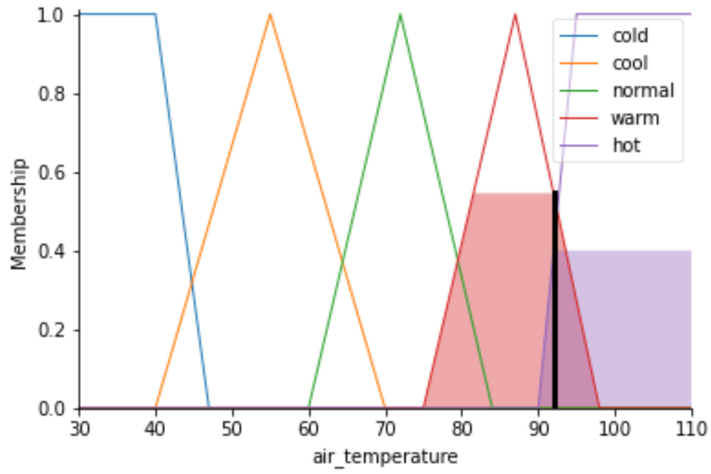
In [14]:
```
# Run simulator
# Evaluate the control system
sprinkler_simulator.compute()
```

In [15]:
```
# Get output value
watering_duration_value = sprinkler_simulator.output['watering_duration']
```

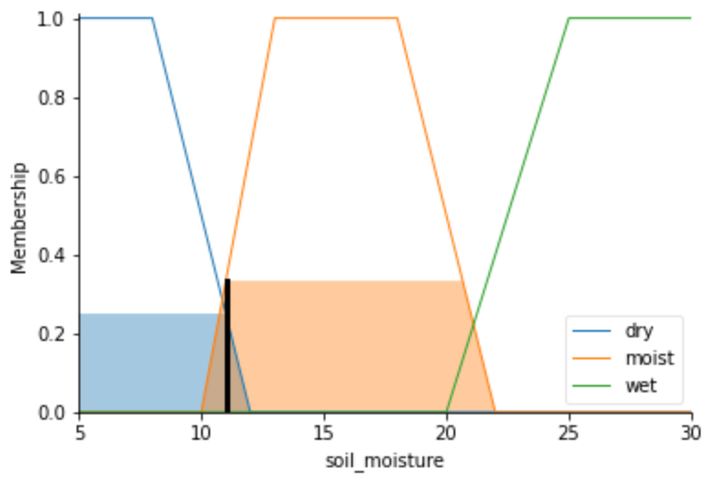
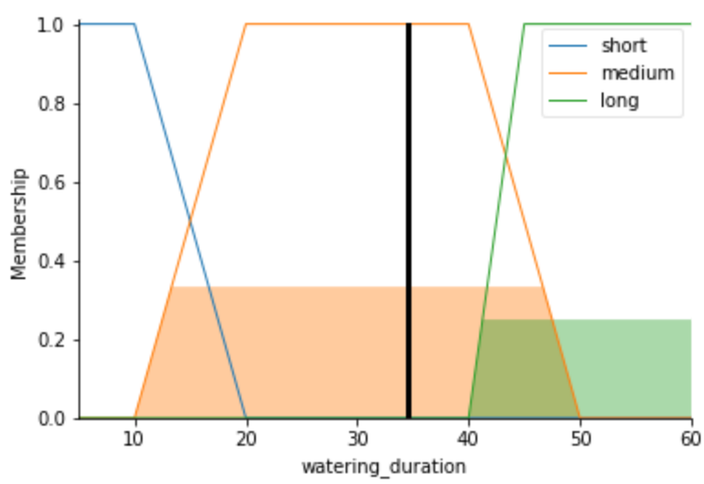
In [16]:
```
# Print output value
print(f"The watering duration is {round(watering_duration_value, 1)}%")
```

```
The watering duration is 34.6%
```

In [17]:
```
# Display fuzzy membership functions
# Visualize membership functions (optional)
air_temperature.view(sprinkler_simulator) # 92
```



In [18]:
```
soil_moisture.view(sprinkler_simulator) # 11
```



In [19]:
```
watering_duration.view(sprinkler_simulator) # 34.6
```